

Telomeres and Cellular Senescence

Sérgio F genetic report
2 of October 2026 (local time)

Name	Sérgio F
Focus	longevity and immortality (84 aging mechanisms)
Data	Crude DNA (chip) + local imputation (only DR2 ≥ 0.8)
Content	7 Curated SNPs · 154 loci of GWAS · 3 polygenic scores · 2 861 pathogenic variants of ClinVar in 26 genes

What is: The genetic predisposition of Sérgio F for telomere length (97 loci of GWAS and a 87 variant score), the TERT locus (telemer x cancer compromise), the senescence genes (9p21/CDKN2A-B, p21, IL-6 of the secretory phenotype) and the acceleration of epigenetic age, more rare variants of telomere diseases.
Privacy: the analysis was done locally; the genotypes never left the computer. Public databases were only sent SNP identifiers.

Important: genetic predispositions, not a diagnosis. The conclusions come only from the genes (without analysis or exams). All practical ideas are suggestion (not applied), to decide with the doctor.

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1. Abstract: favourable and unfavourable

Favourable	Unfavourable
• TERT rs2736100 CC: 2 copies of the allele that extends the telomeres (commitment: see unfavourable). • 9p21 (CDKN2B rs1063192 AA): without the G allele of more coronary disease in the senescence locus p16/p15. • 19 loci of favourable telomeres, among them RTEL1, POT1, ACYP2 and ZNF208 (genes of telomeres maintenance). • Epigenetic age: Typical score (the 59th percentile); 8 loci favourable. • ClinVar: no pathogenic variant in the 26 genes of telomere diseases (TERT, TERC, DKC1, RTEL1, PARN...).	• Genetically shorter telomeres: 87 score at the 25th percentile and classical 7 score SNPs at the 24th percentile. • TERT rs10069690 TT (2 copies of the shorter telomere allele) and TERC rs10936599 TC. • TERT rs2736100 CC (commitment): The same allele that extends telomeres increases the risk of glioma, lung adenocarcinoma and myeloproliferative neoplasms. • IL6 –174 GG: genotype of more IL-6 in some studies (secretory phenotype of senescent cells). • 20 loci of unfavourable telomeres (e.g. ACTRT3, MPHOSPH6, TERF2).

Reading for immortality: The genetic telomeric reserve is slightly below the average (~the 25th percentile), with TERT pulling in both directions. For immortality, the priority is not to force telomerase (TERT cancer risk alleles discourage it) but to reduce wear: less inflammation and oxidative stress, exercise, sleep, and eliminate senescent cells safely (senolytic pulse of the plan, rapamycin already underway).

2. Method, chain and trust

- **Genotypes:** 27 read directly on the chip (GRCh38, strand +); 109 locally imputed (statistical imputation, full genome reference panel 30x, only DR2 ≥ 0.8); 25 without data ("SNP not found in the data").
- **Strand:** All alleles were aligned to the GRCh38 strand + and verified against alleles and frequencies of the reference panel. 3 alleles of the literature were in the strand coding – and were converted (complement); 24 palindromic SNPs (A/T or C/G) were resolved by the allelic frequency; the non-resolvable palindromic SNPs were out of the scores.
- **Classification:** compares the copies of the effect allele with the average of the reference population (2 × frequency). Difference ≥ 0.5 copies → Favourable or Unfavourable; lower → Typical. Not having a rare variant of risk, or a common variant of strong effect, counts as Favourable. An allele with opposite effects (e.g. longer telomeres but higher risk of cancer) appears as Commitment in both lists.
- **Trust:** chip > imputation (DR2 ≥ 0.8; estimated accuracy of ~95-99% in common variants, lower in rare ones). Evidence: Strong (replicated in large/functional GWAS), Moderate, Weak (small or inconsistent studies).
- **Polygenic scores:** weighted sum of alleles (published coefficients); percentile calculated against the reference population with exactly the same variants.
- **Sources of interpretation:** dbSNP/Ensembl, ClinVar, GWAS Catalog, PGS Catalog and articles cited in the text.

3. Polygenic scores

Score	Percentile	Variants used	Interpretation	Note
Telomere length (87 loci, Liu 2024) G_TL · calculated in this report	25	86 of 87 (99%)	Unfavourable	Low percentile : genetically shorter telomeres (higher reserve lost earlier; lower risk of some cancers).
Telomere length (Classical 7 SNPs, Codd 2013) CUSTOM_TL7 · previous genomic imputation (same method)	24	7 of 7 (100%)	Unfavourable	Confirms the major score.
Epigenetic age acceleration (IEAA, GrimAge, PhenoAge; 39 loci) G_EPI · calculated in this report	59	39 of 57 (68%)	Typical	Typical. Coverage of 37% coefficients: weak reading.

the 50th percentile : median of the reference population. For risk scores, high percentile = higher risk; for "pro" scores (e.g. longer telomeres, longevity), high percentile = more of the trait. Typical = percentile between 30 and 70. The scores explain only a small part of the variation and are not diagnostic.

4. Favourable

Gene/score	SNP	Genotype	Source · evidence	Effect
TERT	rs2736100 Intron 2 (TERT)	CC 2× C	chip Strong	Allele C: longer telomeres (more replicative reserve). Mechanisms: Telomeres and Hayflick Limit; Exhaustion of Stem Cells; Genomic instability
CDKN2B-AS1	rs1063192 3UTRCDKN2B (9p21)	AA 0× G	chip Moderate	Without the risk allele G. 9p21 locus (p16INK4a/p15INK4b/ANRIL), central in senescence. The G allele is associated with more coronary disease and glaucoma. Mechanisms: Cell Senescence; Vascular Aging
RTEL1-TNFRSF6B, LINC03043, ACYP2, ZNF208, ATP8B4, SHQ1	Length of leukocyte telomeres 19 of 97 loci		GWAS GWAS	19 loci in which the genotype of Sérgio F deviates from the average of the reference population in the favourable sense (favourable = longer telomeres). Mechanisms: Telomeres and Hayflick Limit; Exhaustion of Stem Cells
AKIRIN1, TMEM121B, SMARCC1, TRIM59-IFT80, RPL30P12, POLR1HASP	Epigenetic age acceleration (IEAA, GrimAge, PhenoAge) 8 of 57 loci		GWAS GWAS	8 In which the genotype of Sérgio F deviates from the mean of the reference population in the favourable sense (favourable = less accelerated epigenetic age). Mechanisms: Epigenetic Derivative; Epigenetic Changes

5. Unfavourable

Gene/score	SNP	Genotype	Source · evidence	Effect
TERT	rs2736100 Intron 2 (TERT)	CC 2× C	chip Strong	Allele C: increased risk of glioma, lung adenocarcinoma and myeloproliferative neoplasms (the same signal that extends telomeres). Mechanisms: Telomeres and Hayflick Limit; Exhaustion of Stem Cells; Genomic instability
TERT	rs10069690 Intron 4 (TERT)	TT 2× T	chip Moderate	2 copies of the risk allele T. The T allele creates a splicing variant that reduces functional telomerase; it is associated with shorter telomeres and breast and ovarian cancer. Mechanisms: Telomeres and Hayflick Limit; Genomic instability
TERC	rs10936599 MYNN/TERC region	TC 1× T	chip Strong	1 copy of risk allele T. The T allele is associated with shorter telomeres (TERC locus replicated in several GWAS). Mechanisms: Hayflick's Telomeres and Limit
IL6	rs1800795 -174G>C (SASP)	GG 2× G	chip Weak	2 copies of the risk allele G. The G allele is associated, in some studies, with more IL-6 (main component of the secretory phenotype of senescent cells). Mechanisms: Cell Senescence; Inflammaging; Chronic Inflammation
polygenic	G_TL Telomere length (87 loci, Liu 2024)	the 25th percentile	86 variants Score	Low percentile : genetically shorter telomeres (higher reserve lost earlier; lower risk of some cancers). Mechanisms: Telomeres and Hayflick Limit; Exhaustion of Stem Cells
polygenic	CUSTOM_TL7 Telomere length (Classical 7 SNPs, Codd 2013)	the 24th percentile	7 variants Score	Confirms the major score. Mechanisms: Hayflick's Telomeres and Limit
ACTRT3, MPHOSPH6, AP4B1, TERF2, DDX19A, MGA	Length of leukocyte telomeres 20 of 97 loci		GWAS GWAS	20 loci in which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = longer telomeres). Mechanisms: Telomeres and Hayflick Limit; Exhaustion of Stem Cells
CARMIL1, RPN1, TERT, CBX3P9, DIP2C, ZFHX3	Epigenetic age acceleration (IEAA, GrimAge, PhenoAge) 6 of 57 loci		GWAS GWAS	6 In which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = less accelerated epigenetic age). Mechanisms: Epigenetic Derivative; Epigenetic Changes

6. Rare variants (ClinVar) and chip boundaries

All 2 861 variants classified as pathogenic or probably pathogenic in the ClinVar were searched for in the 26 genes of the panel of this report (ACD, BLM, CDKN2A, CDKN2B, CTC1, DCLRE1B, DKC1, LMNA, NAF1, NHP2, NOP10, PARN, POT1, RTEL1, STN1, TERC, TERF1, TERF2IP, TERT, TIF2, WRAP53, WRN, ZCCHC8, ZMPSTE24). Of these, 258 have probes on the chip and additional 4 are in the imputation panel.

Gene	Variant	Genotype	Condition (ClinVar)	Evaluation
CDKN2B	rs1063192 9:22003368 G>A	AA	Three Vessel Coronary Disease; Malignant tumor of breast	Common variant (allele A in about 58% of the chromosomes of the reference population), registered in ClinVar with weak assertion and as "protective". It is not pathogenic: allele A is associated with less coronary disease. Favourable.

Chip limit: the chips genotype well common variants, but often fail in the rare ones (false positives and false negatives; Weedon 2021). The absence of pathogenic variants here does not exclude hereditary syndromes. Only clinical sequencing (panel, exome or genome) in a certified laboratory does so.

7. GWAS loci

For each study: independent loci (one SNP per region of 500 kb, the most significant, $p \leq 5 \times 10^{-8}$), with the genotype of Sérgio F and the classification against the reference population. The typical loci are in the complete table (CSV); here we show those that move away from the average.

Length of leukocyte telomeres

Liu WS et al. 2024 (large reference cohort, 327 790 people). Effect allele: longer telomeres : favourable. 97 loci: 19 favourable, 20 unfavourable, 50 typical, 8 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
ACTRT3	rs35446936	A ↓	AG	0.24	imputation (DR2=1.00)	Unfavourable
RTKL1-TNFRSF6B, RTKL1	rs35640778	A ↓	GG	0.01	chip	Favourable
MPHOSPH6	rs2303262	T ↑	TC	0.78	chip	Unfavourable
LINC03043 - POT1	rs6968500	C ↑	CC	0.48	imputation (DR2=0.98)	Favourable
ACYP2	rs965109	T ↓	CC	0.04	imputation (DR2=1.00)	Favourable
ZNF208 - ZNF257	rs8105767	G ↑	GG	0.27	imputation (DR2=1.00)	Favourable
AP4B1	rs17464525	A ↓	AG	0.18	imputation (DR2=1.00)	Unfavourable
ATP8B4	rs11857230	T ↓	AA	0.35	imputation (DR2=1.00)	Favourable
SHQ1	rs78491606	C ↓	AA	0.02	chip	Favourable
XPO7	rs10101881	C ↑	CC	0.59	imputation (DR2=0.98)	Favourable
TERF2	rs9939705	C ↑	GG	0.26	imputation (DR2=0.99)	Unfavourable
TK1	rs144204502	T ↓	CC	0.01	chip	Favourable
DDX19A	rs4985525	A ↓	AA	0.40	imputation (DR2=0.96)	Unfavourable
MGA	rs17677991	G ↑	CC	0.38	imputation (DR2=1.00)	Unfavourable
BET1L - RIC8A	rs11245998	G ↓	GG	0.72	imputation (DR2=1.00)	Unfavourable
PML	rs5742915	C ↑	TT	0.44	chip	Unfavourable
TNPO3	rs7783384	C ↑	TT	0.62	imputation (DR2=1.00)	Unfavourable
GATA2-AS1	rs2713576	A ↓	AA	0.43	imputation (DR2=0.98)	Unfavourable
LINC00910	rs56799554	G ↓	GA	0.18	imputation (DR2=0.94)	Unfavourable
TENT4B	rs11076521	T ↑	TT	0.71	imputation (DR2=0.91)	Favourable
RAD54B - VIRMA	rs7829886	A ↓	TT	0.30	imputation (DR2=1.00)	Favourable
UBE2D2	rs113845783	T ↑	TC	0.07	imputation (DR2=0.98)	Favourable
PPRC1	rs4919611	A ↑	AC	0.85	imputation (DR2=0.98)	Unfavourable
FRG2FP - TUBB8P8	rs79583340	A ↓	TT	0.05	imputation (DR2=0.90)	Favourable
MXD4	rs2282764	G ↓	GA	0.16	imputation (DR2=1.00)	Unfavourable
SPOCD1 - PTP4A2	rs61779657	G ↑	AA	0.39	imputation (DR2=1.00)	Unfavourable
CASTOR3P	rs151024111	T ↓	TC	0.22	imputation (DR2=0.99)	Unfavourable
SMC1B	rs6006995	C ↑	AA	0.37	imputation (DR2=0.98)	Unfavourable
SKIL	rs116729888	T ↓	CC	0.04	imputation (DR2=0.95)	Favourable
SLC7A10 - CEBPA	rs12151289	C ↓	GG	0.03	chip	Favourable
CLIP1 - ZCCHC8	rs10846920	T ↑	TT	0.74	imputation (DR2=0.98)	Favourable
PPP1R36	rs11622168	C ↓	GG	0.44	imputation (DR2=0.99)	Favourable

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
SPMIP7 - IKZF1	rs117989553	C ↓	CT	0.07	imputation (DR2=1.00)	Unfavourable
KLF7P1 - RNU1-70P	rs74370602	C ↓	TT	0.02	imputation (DR2=0.96)	Favourable
BCAR4	rs12924329	T ↑	CC	0.64	imputation (DR2=0.99)	Unfavourable
DEFB125 (intergenic)	rs6054257	A ↑	AG	0.80	imputation (DR2=0.93)	Unfavourable
PRAM1	rs2396106	G ↑	TT	0.36	imputation (DR2=0.96)	Unfavourable
ASB3	rs2009144	A ↑	AA	0.17	imputation (DR2=1.00)	Favourable
ZSCAN5A	rs34187696	A ↓	CC	0.28	imputation (DR2=0.88)	Favourable

Epigenetic age acceleration (IEAA, GrimAge, PhenoAge)

McCartney DL et al. 2021 (34 467 people). Effect allele: more accelerated epigenetic age = unfavourable. 57 loci: 8 favourable, 6 unfavourable, 25 typical, 18 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
CARMIL1 - SCGN	rs10447389	A ↓	GG	0.28	chip	Unfavourable
RPN1	rs1126828	G ↓	GA	0.78	chip	Unfavourable
AKIRIN1	rs12043492	T ↑	CC	0.45	imputation (DR2=1.00)	Favourable
TMEM121B, LINC01664	rs75243280	C ↑	TT	0.28	imputation (DR2=0.93)	Favourable
TERT	rs2736100	A ↓	CC	0.49	chip	Unfavourable
SMARCC1	rs79111787	C ↑	TT	0.03	chip	Favourable
CBX3P9 - KRT18P22	rs2858830	G ↓	AA	0.35	imputation (DR2=1.00)	Unfavourable
TRIM59-IFT80, TRIM59	rs1047210	C ↑	AA	0.43	imputation (DR2=0.99)	Favourable
RPL30P12 - TMEM263-DT	rs10735418	C ↓	CC	0.39	imputation (DR2=0.99)	Favourable
POLR1HASP	rs28780071	T ↑	CC	0.35	imputation (DR2=1.00)	Favourable
Y_RNA - PERPP1	rs9853301	G ↑	GT	0.88	imputation (DR2=0.96)	Favourable
DIP2C - LARP4B	rs11253338	T ↑	TC	0.19	imputation (DR2=0.98)	Unfavourable
MSI2	rs116853700	A ↑	GG	0.03	imputation (DR2=0.98)	Favourable
ZFH3	rs34003787	T ↑	TC	0.09	imputation (DR2=0.87)	Unfavourable

8. Map for 84 aging mechanisms

Mechanism	Favourable	Unfavourable	Interpretation
Hayflick's Telomeres and Limit	TERT rs2736100; Length of leukocyte telomeres	TERT rs2736100; TERT rs10069690; TERC rs10936599; G_TL; CUSTOM_TL7; Length of leukocyte telomeres	Mild disadvantage: scores in the ~the 25th percentile; TERT in commitment.
Exhaustion of Stem Cells	TERT rs2736100; Length of leukocyte telomeres	TERT rs2736100; G_TL; Length of leukocyte telomeres	Shorter telomeres may limit the replicative reserve of stem cells (see the stem cell report).
Genomic instability	TERT rs2736100	TERT rs2736100; TERT rs10069690	The unfavourable predominates
Cell Senescence	CDKN2B-AS1 rs1063192	IL6 rs1800795	9p21 favourable; IL6GG mild unfavourable; p21 informative.
Vascular Aging	CDKN2B-AS1 rs1063192	—	Predominant
Epigenetic Derivative	Epigenetic age acceleration (IEAA, GrimAge, PhenoAge)	Epigenetic age acceleration (IEAA, GrimAge, PhenoAge)	Typical age (weak reading).
Epigenetic Changes	Epigenetic age acceleration (IEAA, GrimAge, PhenoAge)	Epigenetic age acceleration (IEAA, GrimAge, PhenoAge)	No relevant deviation.
Inflammaging	—	IL6 rs1800795	IL6 –174 GG.
Chronic Inflammation	—	IL6 rs1800795	The unfavourable predominates

9. Practical section — suggestion (not applied)

Coherent with the existing genetic plan and the current stack reported. Nothing was applied here; everything is to discuss with the doctor.

#	Suggestion	Why (genes)	State
1	Aerobic exercise and regular resistance	It is associated with longer telomeres and less senescent cells; it compensates for a lower genetic reserve.	suggestion (not applied)
2	Omega-3 to 3.2-4 g/day of EPA+DHA (already taking fish oil)	High levels of omega-3 were associated with shorter telomeres (Farzaneh-Far 2010).	suggestion (not applied)
3	7-9 sleep and chronic stress control	Short sleep and chronic stress are associated with shorter telomeres.	suggestion (not applied)
4	Senolytic pulse only with fisetin + quercetin and only with normal ALT (as in the plan)	It eliminates senescent cells; pause the quercetin outside the pulse while the ALT is high is a suggestion (not applied).	suggestion (not applied)
5	Keep rapamycin 4 mg of 6 in 6 weeks (already in the stack)	MTORC1 feeds the secretory phenotype of senescent cells; rapamycin reduces this phenotype. Without changing the dose.	suggestion (not applied)
6	More fiber and Mediterranean diet	Less chronic inflammation (IL6 GG) and more butyrate.	suggestion (not applied)
7	Maintaining vitamin D3 as it is in the plan (10 000 IU)	No conflict with the genes of this report.	suggestion (not applied)
8	Measurement of telomere length by Flow-FISH (optimal, once)	Confirms if the genetic predisposition translates into short telomeres.	suggestion (not applied)

Avoiding or with caution

- **Unvalidated telomerase activators** (TA-65, astragaloside, cycloastragenol, astragalus extracts) and epitalon (the plan suggests to stop it; suggestion not applied): with TERT rs2736100 CC and other alleles TERT at risk of cancer, activate telomerase may favor malignant clones. (suggestion, not applied)
- Growth hormone or IGF-1 secretagogues without medical indication. (suggestion, not applied)

10. Limitations

- The telomere score predicts the length in leukocytes; the real length also depends on age, lifestyle and tissue.
- The TERT locus has opposite effects (telomers vs. cancer) and should not be read as "good" or "bad".
- The real epigenetic age is only measured with a blood methylation clock.
- The GWAS loci show average associations in the population; the "mapped" gene is the closest and not always the causal gene.
- The effects of isolated SNPs are small; several studies with only one population. They do not allow clinical conclusions by themselves.
- Imputation: an imputed genotype may be wrong, especially in rare variants; the imputed ones are marked.
- This report uses only genes: analysis, exams or measurements were not used.

11. Full table SNP to SNP (cured SNPs)

Gene	SNP	Effect allele (+)	Genotype	Freq.	Source	Chain	Class	Evidence
TERT	rs2736100 Intron 2 (TERT)	C (commitment)	CC	0.506	chip	strand + (without conversion)	Commitment	Strong
TERT	rs7705526 Intron 2 (TERT)	A (commitment)	AC	0.346	chip	strand + (without conversion)	Typical	Strong
TERT	rs10069690 Intron 4 (TERT)	T (risk)	TT	0.281	chip	strand + (without conversion)	Unfavourable	Moderate
TERC	rs10936599 MYNN/TERC region	T (risk)	TC	0.240	chip	strand + (without conversion)	Unfavourable	Strong
CDKN2B-AS1	rs1063192 3UTRCDKN2B (9p21)	G (risk)	AA	0.424	chip	strand + (without conversion)	Favourable	Moderate
CDKN1A	rs1801270 Ser31Arg (p21)	A (informative)	AC	0.065	chip	strand + (without conversion)	Information	Weak
IL6	rs1800795 -174G>C (SASP)	G (risk)	GG	0.575	chip	palindromic: confirmed by frequency (reference population 0.58)	Unfavourable	Weak

Notes about the typical, informative and without data: TERT rs7705526 (AC): Allele A: longer telomeres. CDKN1A rs1801270 (AC): Variant of p21 (stopping cycle and senescence); inconsistent associations.